中文文档

巴适声工厂 · 隐私版 (Bashi Voice Factory Privacy Edition)

License MIT version 0.1.0 python 3.12 embed platform Windows

版本： 0.1.0

完全离线运行的本地语音工厂网页应用。首次启动联网下载完依赖与模型之后，所有文字转语音、语音转文字、音频生成、文件保存都在你自己的电脑上完成，没有任何音频数据上传云端。

> ⚡ 想要云端语音质量、更快的合成速度？请关注 巴适声工厂 · 极速版 (Bashi Voice Factory Turbo) — 微软 Edge TTS 引擎、14 种语言、5 万字长文。需要联网。

作者： Alex Li (ncorecpu@gmail.com) 许可： MIT License 源码： https://github.com/gtree965/bashi-voice-factory-privacy 下载： GitHub Releases · files.fm 镜像

✨ 核心亮点

🔒 完全本地文字转语音

•Qwen3-TTS-12Hz-1.7B-CustomVoice 本地推理 — 合成过程零云端调用。
•9 个精选音色，内置母语试听样例。
•10 种语言：中文（普通话、粤语、四川话、北京话、东北话）、英文（美/英）、日语、韩语、德语、法语、俄语、葡萄牙语、西班牙语、意大利语。
•自然语言指令控制 — 风格、口音、情绪都可通过 instruct 提示调整。

🎯 后端自动选择

•GGUF + Vulkan：AMD / Intel / 集成显卡用户默认路线（速度快、内存友好）。
•PyTorch + CUDA：NVIDIA 用户进阶路线（权重需单独下载）。
•CPU 回退：入门级硬件（如 Intel N100 系列）也能跑。
•一键 测速 校准 ETA 估算，让等待时间贴合你这台机器。

🎙 本地离线音视频转文字 (STT)

•SenseVoice Small（默认，242 MB）— 中英日韩粤多语种，非自回归架构，CPU 上速度快。
•Parakeet TDT 0.6B（可选，661 MB）— 英文专用，NVIDIA 出品，WER 约 1.7%。
•Silero VAD 精准分段 — 时间戳准确、零重叠、零卡顿。
•实时滚屏字幕 SSE 流式输出；支持导出 TXT、SRT、VTT。

🛡 智能首次下载，断点续传

•HTTP Range / resume 内置 — 一次 WiFi 闪断不会让 2.2 GB 的 GGUF 下载从零开始。
•3 次重试，指数退避 — pip 依赖与模型文件都有同样的兜底逻辑。
•国内友好镜像：自动识别中国时区使用阿里云 PyPI；GGUF 运行模型走 ModelScope；STT 模型走 hf-mirror.cn。

📱 局域网共享，手机平板可访问

•首次启动询问是否绑定 `0.0.0.0`，允许同一 WiFi 下其他设备访问。

•手机浏览器输入电脑的局域网 IP 即可使用，跟本地网页一样流畅。

🚀 快速上手

最简流程

1.把下载的 zip 解压到任意文件夹（桌面或随便哪里都行）。

2.在解压后的顶层目录里 双击 `Start_启动.bat`。

3.首次启动会：

•安装 Python 依赖（约 700 MB，根据网速 / CPU 不同需要 8-15 分钟）

•询问是否下载 GGUF 模型（约 2.2 GB，2-15 分钟）

•自动在默认浏览器打开 `http://127.0.0.1:5050`
4.首次完成后，之后每次启动都是离线，5 秒左右即可打开。

进阶

如果想跳过顶层启动器，可以直接运行 `bashi-privacy-app\run_portable.bat`，效果完全一样。

🌐 网络行为说明

隐私承诺：本程序不主动联网、不上传音频、不收集使用数据。程序永远不会自动检查任何在线资源。 所有联网行为均在下文有完整说明，且需要用户主动触发 — 无后台轮询、无统计分析、无静默更新探测。

仅首次启动（一次性，约 700 MB + 约 2.2 GB）：

•pip 依赖来自 `https://mirrors.aliyun.com/pypi/simple/`（海外用户默认 PyPI）
•GGUF 模型来自 `https://modelscope.cn/models/gtree592/bashi-qwen3-tts-1.7b-customvoice-gguf-runtime`
•STT 模型（用户主动点击下载）来自 `https://hf-mirror.com/csukuangfj/...`（备用 HuggingFace）

运行时网络行为：零。文字转语音、音视频转文字、音频导出全部本机完成。

手动更新检查（仅用户主动触发，永不自动）：UI 右下角"查看最新版本"按钮在新浏览器标签页打开发布页。也可以随时手动访问：

•GitHub Releases: https://github.com/gtree965/bashi-voice-factory-privacy/releases
•files.fm 镜像: https://files.fm/u/juvstxmrez
离网 / 敏感场景部署（银行、医疗、政务、企业内网）：在联网机器上完成首次启动的所有下载后，整个 `bashi-voice-factory-privacy-v0.x.0/` 解压文件夹可以原样拷贝到离网/物理隔离机器上运行，之后所有功能

均无需任何网络访问。程序没有任何遥测端点需要防火墙拦截，没有自动更新守护进程，也没有回调 URL。可用 Wireshark / 资源监视器在正常使用过程中监测出站流量验证 — 应当全程静默。

💻 硬件要求

下表是作者实测的数据。首次运行后点右侧面板的"测速"按钮，程序会根据你这台机器实际表现重新校准 ETA 估算 — 不用照搬此表。

硬件	自动选择的后端	"你好。" 探测（25 字）	1,000 字合成预计	状态
AMD RX 9060 XT (16 GB)	GGUF + Vulkan	约 3 秒	几分钟	✓ 2026-05 实测
AMD RX 590 (8 GB)	GGUF + Vulkan	3-5 秒	约 5-10 分钟	√ 实测
Intel N305 笔记本 + UHD 集显	GGUF + Vulkan / DirectML	53 秒	25-46 分钟	✓ 2026-05-25 实测
Intel N100 迷你主机 + UHD 集显	GGUF + Vulkan / DirectML	126 秒	58 分钟 - 1 小时 49 分	✓ 2026-05-25 实测
NVIDIA RTX / Apple Silicon / Intel Arc	—	暂未实测	暂未实测	v0.1 未验证

> ⚠ 入门级 CPU（Intel N100 / N305 一类）仅适合短句试用 — 单次生成建议不超过 200 字。 这类机器跑 5,000 字长文需要 2-9 小时。如果想做长文音频（讲座、有声书），请用独立显卡硬件（AMD RX 500/600/9000 系、NVIDIA RTX 类）。
> ℹ Windows 上的 NVIDIA 用户也可以走 GGUF + Vulkan 路线（NVIDIA 驱动支持 Vulkan）。PyTorch + CUDA 路线在 v0.1 需要手动配置权重，不会自动启用。

🧩 硬件兼容性矩阵

除了上表里具体测过的几台机器，下表说明主流 Windows 硬件在自动后端选择下走的加速路径：

硬件类别	实际加速路径	状态
NVIDIA RTX 30 / 40 / 50	GGUF + Vulkan + DirectML	⚠ 能跑但**次优** — v0.1 未走 CUDA
NVIDIA GTX 10 / 16	GGUF + Vulkan + DirectML	⚠ 同上
AMD RX 500 / 600 / 7000 / 9000（独立显卡）	GGUF + Vulkan + DirectML	✅ 实测（RX 590、RX 9060 XT）
Intel Arc A 系列（A380 / A580 / A750 / A770）	GGUF + Vulkan + DirectML	✅ 理论可用、未实测
Intel 集显（UHD / Iris Xe / Arc iGPU）	GGUF + Vulkan + DirectML	✅ 实测（Intel N305 UHD）

硬件类别	实际加速路径	状态
AMD APU 集显（Vega 7/8、RDNA 2/3）	GGUF + Vulkan + DirectML	✅ 理论可用、未实测
纯 CPU（无可用 GPU / 无驱动）	GGUF + CPU（ggml-cpu 自动 SIMD）	✅ 可用；仅在无 GPU 时自动选择
NPU（Snapdragon X / Lunar Lake / Ryzen AI 300）	—	❌ 暂未利用
ARM64 Windows（Snapdragon X Copilot+ PC）	—	❌ 暂不支持（走 x64 仿真较慢）

⚙️ 已知限制（路线图）

•NVIDIA 独显用户当前走 Vulkan、未走 CUDA。 RTX 4090 跟 N100 集显走同一条路。独立的 NVIDIA + CUDA 构建在 v0.2 路线图。
•弱集显可能比纯 CPU 还慢。 Intel N100 类硬件上，DirectML / Vulkan 的数据传输与调度开销可能盖过 GPU 收益。可在 cmd 窗口里 `set GGUF_LLM_USE_GPU=0 && set GGUF_ONNX_PROVIDER=CPU` 后再启动来 A/B 测试，欢迎反馈实测数字。
•NPU 加速尚未利用（Snapdragon X Elite、Intel Lunar Lake AI Boost、AMD Ryzen AI 300）。v0.2+ 调研中。
•ARM64 Windows 暂未原生支持（Surface Pro 11、Galaxy Book4 Edge 等）。x64 仿真能跑但速度受限。原生 ARM64 构建是 v0.3+ 候选。

🐛 常见问题排查

现象	可能原因 / 处理
双击 `Start_启动.bat` 报 "Array index expression is missing"	旧版 zip。请从 files.fm 重新下载最新版；该 BOM 问题在 v0.1.0 正式版已修复
pip 安装途中 WiFi 闪断后中止	自动重试 3 次（5/30/120 秒退避）。3 次都失败时，修好网络后重新运行启动器即可（pip 会跳过已装好的包）
GGUF 下载中断	同样自动重试，且支持 HTTP Range 续传，重新运行启动器从 `.part` 文件续传
启动报 "No usable backend was found"	查 `launch_log.txt`。常见原因：GGUF 运行 DLL 缺失、显卡驱动过旧、可用内存 <8 GB。程序会打印中英双语提示告知具体原因
STT 下载闪过 "镜像失败"	v0.1.0 正式版不会出现 — 是旧版残留的 ModelScope 路径，已删除

完整日志路径： `bashi-privacy-app\launch_log.txt`

📦 zip 包目录结构

```
bashi-voice-factory-privacy-v0.1.0/
```

```
├── Start_启动.bat                                    ← 双击这里
├── README.md                                          ← 英文文档
├── README_CN.md                                       ← 本文件
├── LICENSE                                            ← MIT 许可证
├── VERSION                                            ← 发布版本号
├──── 巴适声工厂隐私版使用手册_Bashi_Voice_Factory_Privacy_Edition_User_Guide.pdf
|                                                  ← 中英双语帮助 PDF
├── bashi-privacy-app/                                 ← 程序代码 + 嵌入式 Python
│   ├── run_portable.bat                               ← 等效启动器（直接路径）
│   ├── README.md                                      ← 项目说明副本
│   └── ...
└── vulkan_backend_spike/                              ← GGUF 运行时（首次启动自动填
充）
    └── Qwen3-TTS-GGUF/
```

📄 许可

MIT License © 2026 Alex Li

第三方组件保留各自原始许可：

•Qwen3-TTS-12Hz-1.7B-CustomVoice：通义实验室（阿里巴巴）Apache 2.0
•GGUF runtime：基于 HaujetZhao/Qwen3-TTS-GGUF
•SenseVoice Small / Silero VAD / Parakeet TDT：详见各自模型卡

👤 作者

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欢迎通过 GitHub release 页面或邮件提交 issue、反馈或功能建议。

English

Bashi Voice Factory Privacy Edition (巴适声工厂 · 隐私版)

License MIT version 0.1.0 python 3.12 embed platform Windows

Version: 0.1.0

A fully offline desktop web app for high-quality text-to-speech and speech-to-text. After the one-time first-launch download, everything runs on your own machine — TTS synthesis, audio export, transcription, and storage. No audio data ever leaves your computer.

> ⚡ Need cloud-quality voices and faster generation? Check out Bashi Voice Factory Turbo (巴适声工厂 · 极速版) — Microsoft Edge TTS, 14 languages, 50,000-character long text. Requires internet.

Author: Alex Li (ncorecpu@gmail.com) License: MIT License Source code: https://github.com/gtree965/bashi-voice-factory-privacy Download: GitHub Releases · files.fm mirror

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✨ Highlight Features

🔒 Fully Local Text-to-Speech

•Local Qwen3-TTS-12Hz-1.7B-CustomVoice running entirely on your machine — no cloud calls during synthesis.
•9 curated voice presets with native-language sample previews built in.
•10 languages supported: Chinese (Mandarin, Cantonese, Sichuanese, Beijing, Northeast), English (US/UK), Japanese, Korean, German, French, Russian, Portuguese, Spanish, Italian.
•Instruction control for style, accent, emotion via natural-language prompts.

🎯 Automatic Backend Selection

•GGUF + Vulkan for AMD / Intel / iGPU users (default, fast, RAM-friendly).
•PyTorch + CUDA for NVIDIA users (advanced setup; weights download separately).
•CPU fallback for entry-level hardware (works on Intel N100-class boxes).
•One-click speed test calibrates ETA estimates to your specific machine.

🎙️ Local Offline Speech-to-Text (STT)

•SenseVoice Small (default, 242 MB) — multilingual zh/en/ja/ko/yue, non-autoregressive (fast on CPU).
•Parakeet TDT 0.6B (optional, 661 MB) — English specialist, NVIDIA's ~1.7% WER champion.
•VAD-based segmentation via Silero — accurate timestamps, no chunk-boundary stutter.
•Live transcript streaming via SSE; exports to TXT / SRT / VTT.

🛡️ Smart First-Launch Download with Resume

•HTTP Range / resume built in — a transient WiFi drop won't restart your 2.2 GB GGUF download from zero.
•3-attempt retry with exponential backoff for both pip dependencies and model files.
•China-friendly mirrors: Aliyun PyPI mirror auto-detected by locale; GGUF runtime hosted on ModelScope; STT models hosted on hf-mirror.cn.

📱 LAN Sharing for Phone / Tablet Use

•First-launch prompt: bind to `0.0.0.0` to allow access from any device on the same WiFi.

•Compatible with mobile browsers — type the LAN IP and use it like a local web app.

🚀 Quick Start

Easiest path

1.Unzip the downloaded file to any folder (Desktop or wherever you like).

2.Double-click `Start_启动.bat` at the top level of the extracted folder.

3.First launch will:

•Install Python dependencies (~700 MB, 8-15 min depending on network/CPU)

•Prompt to download the GGUF model (~2.2 GB, 2-15 min depending on network)

•Open `http://127.0.0.1:5050` in your default browser

4.After first launch, all subsequent launches are offline and start in ~5 seconds.

Advanced

Users who prefer to bypass the top-level launcher can run `bashi-privacy-app\run_portable.bat` directly. Both paths are identical.

🌐 Network Behavior

Privacy commitment: this app does not phone home, does not upload audio, does not collect telemetry. The app never automatically checks any internet resource. Every network operation is documented below and requires explicit user action — no background polling, no analytics, no silent update probes.

First launch only (one-time, ~700 MB + ~2.2 GB):

•pip dependencies from `https://mirrors.aliyun.com/pypi/simple/` (or PyPI default outside China)
•GGUF model from `https://modelscope.cn/models/gtree592/bashi-qwen3-tts-1.7b-customvoice-gguf-runtime`
•STT model (when user opts in) from `https://hf-mirror.com/csukuangfj/...` (or HuggingFace fallback)

Runtime network activity: zero. TTS synthesis, STT transcription, and audio export are 100% on-device.

Manual update check (user-initiated only, never automatic): the UI footer "Check for updates" button opens a release page in a new browser tab. You can also check manually anytime at:

•GitHub Releases: https://github.com/gtree965/bashi-voice-factory-privacy/releases
•files.fm mirror: https://files.fm/u/juvstxmrez
Air-gapped / sensitive deployment (banks, healthcare, government, internal corporate networks): after first-launch downloads complete on a connected machine, the entire `bashi-voice-factory-privacy-v0.x.0/` extracted folder can be copied to an offline / physically-isolated machine and run with zero further network access required. The app has no telemetry endpoints to firewall, no auto-update worker, no callback URLs. To verify, monitor outbound traffic with Wireshark / Resource Monitor during a normal session — it should be silent.

💻 Hardware Expectations

The table below shows actually measured numbers from author hardware. The built-in speed test (click 测速 / Speed Test on the right-hand panel) calibrates the ETA panel to your specific machine on first use, so you don't have to extrapolate from this table.

Hardware	Auto-selected backend	"你好。" probe (25 chars)	1,000-char synthesis ETA	Status
AMD RX 9060 XT (16 GB)	GGUF + Vulkan	~3 s	a few minutes	✓ Tested 2026-05
AMD RX 590 (8 GB)	GGUF + Vulkan	3-5 s	~5-10 min	✓ Tested
Intel N305 laptop + UHD iGPU	GGUF + Vulkan / DirectML	53 s	25-46 min	✓ Tested 2026-05-25
Intel N100 mini-PC + UHD iGPU	GGUF + Vulkan / DirectML	126 s	58 min - 1h49 min	✓ Tested 2026-05-25
NVIDIA RTX / Apple Silicon / Intel Arc	—	not yet measured	not yet measured	not validated in v0.1

> ⚠ Entry-level CPUs (Intel N100 / N305 class) are only practical for short text — under ~200 characters per generation. A 5,000-character essay would take 2-9 hours on these boxes. For long-form audio (lectures, audiobooks), please use discrete-GPU hardware (AMD RX 500/600/9000 series, NVIDIA RTX class).
> NVIDIA users on Windows can also use the GGUF + Vulkan path (NVIDIA cards support Vulkan via the proprietary driver). The PyTorch + CUDA path in v0.1 requires manual weight setup; it is not auto-configured.

🧩 Hardware Coverage Matrix

Beyond the specific machines benchmarked above, here is how the auto-selected acceleration path maps across mainstream Windows hardware:

Hardware class	Acceleration path used	Status
NVIDIA RTX 30 / 40 / 50	GGUF + Vulkan + DirectML	⚠ Works, but **suboptimal** — no CUDA path in v0.1
NVIDIA GTX 10 / 16	GGUF + Vulkan + DirectML	⚠ Same as above
AMD RX 500 / 600 / 7000 / 9000 (discrete)	GGUF + Vulkan + DirectML	✅ Tested (RX 590, RX 9060 XT)
Intel Arc A-series (A380 / A580 / A750 / A770)	GGUF + Vulkan + DirectML	✅ Should work — not yet measured
Intel iGPU (UHD / Iris Xe /	GGUF + Vulkan +	✅ Tested (Intel N305 UHD)

Hardware class	Acceleration path used	Status
Arc iGPU)	DirectML	
AMD APU iGPU (Vega 7/8, RDNA 2/3)	GGUF + Vulkan + DirectML	✅ Should work — not yet measured
CPU only (no usable GPU / no driver)	GGUF + CPU (ggml-cpu auto-SIMD)	✅ Works; auto-selected only when no GPU detected
NPU (Snapdragon X / Lunar Lake / Ryzen AI 300)	—	❌ Not yet utilized
ARM64 Windows (Snapdragon X Copilot+ PC)	—	❌ Not supported (runs via slow x64 emulation)

⚙️ Known Limitations (roadmap)

•NVIDIA discrete GPU users currently run via Vulkan, not CUDA. A 4090 takes the same path as an N100 iGPU. A dedicated NVIDIA + CUDA build is on the v0.2 roadmap.
•Weak iGPU may be slower than pure CPU. On Intel N100-class hardware, DirectML / Vulkan transfer + scheduling overhead can outweigh GPU benefit. A/B test by launching from a cmd window after `set GGUF_LLM_USE_GPU=0 && set GGUF_ONNX_PROVIDER=CPU` — feedback on real speed numbers welcome.
•NPU acceleration is not yet used (Snapdragon X Elite, Intel Lunar Lake AI Boost, AMD Ryzen AI 300). Under investigation for v0.2+.
•ARM64 Windows is not natively supported (Surface Pro 11, Galaxy Book4 Edge, etc.). x64 emulation works but is slow. Native ARM64 build is a v0.3+ candidate.

🐛 Troubleshooting

Symptom	Likely cause / fix
`Start_启动.bat` shows "Array index expression is missing"	Old zip — re-download the latest from files.fm; the BOM bug was fixed in v0.1.0 final
pip install stops mid-way after WiFi blip	Retry triggers automatically (3 attempts, 5s/30s/120s backoff). If all 3 fail, fix network and re-run the launcher — pip caches what's already installed
GGUF download interrupted	Same: retries with HTTP Range/resume; re-run launcher to continue from `.part` file
App exits with "No usable backend was found"	Check `launch_log.txt` — usually GGUF runtime DLL missing, GPU driver outdated, or RAM <8 GB. App now prints a bilingual advisory with specific causes
STT download says "镜像失败"	Should not happen in v0.1.0 final — old behavior from removed ModelScope path

Full log path: `bashi-privacy-app\launch_log.txt`

📦 What's in the Zip

```
bashi-voice-factory-privacy-v0.1.0/
├── Start_启动.bat                              ← double-click here
├── README.md                                   ← this file
├── README_CN.md                                ← Chinese version
├── LICENSE                                     ← MIT license
├── VERSION                                     ← release version
├─── 巴适声工厂隐私版使用手册_Bashi_Voice_Factory_Privacy_Edition_User_Guide.pdf
│                                                       ← help PDF (bilingual)
├── bashi-privacy-app/                          ← app code + embedded
Python
│   ├── run_portable.bat                        ← same launcher, direct
path
│   ├── README.md                               ← project README copy
│   └── ...
└── vulkan_backend_spike/                       ← GGUF runtime (populated
on first launch)
    └── Qwen3-TTS-GGUF/
```

📄 License

Bundled third-party components retain their original licenses:

•Qwen3-TTS-12Hz-1.7B-CustomVoice: Tongyi Lab (Alibaba), Apache 2.0
•GGUF runtime: based on HaujetZhao/Qwen3-TTS-GGUF
•SenseVoice Small / Silero VAD / Parakeet TDT: see respective model cards

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Issues, feedback, and feature requests welcome via the GitHub release page or email.